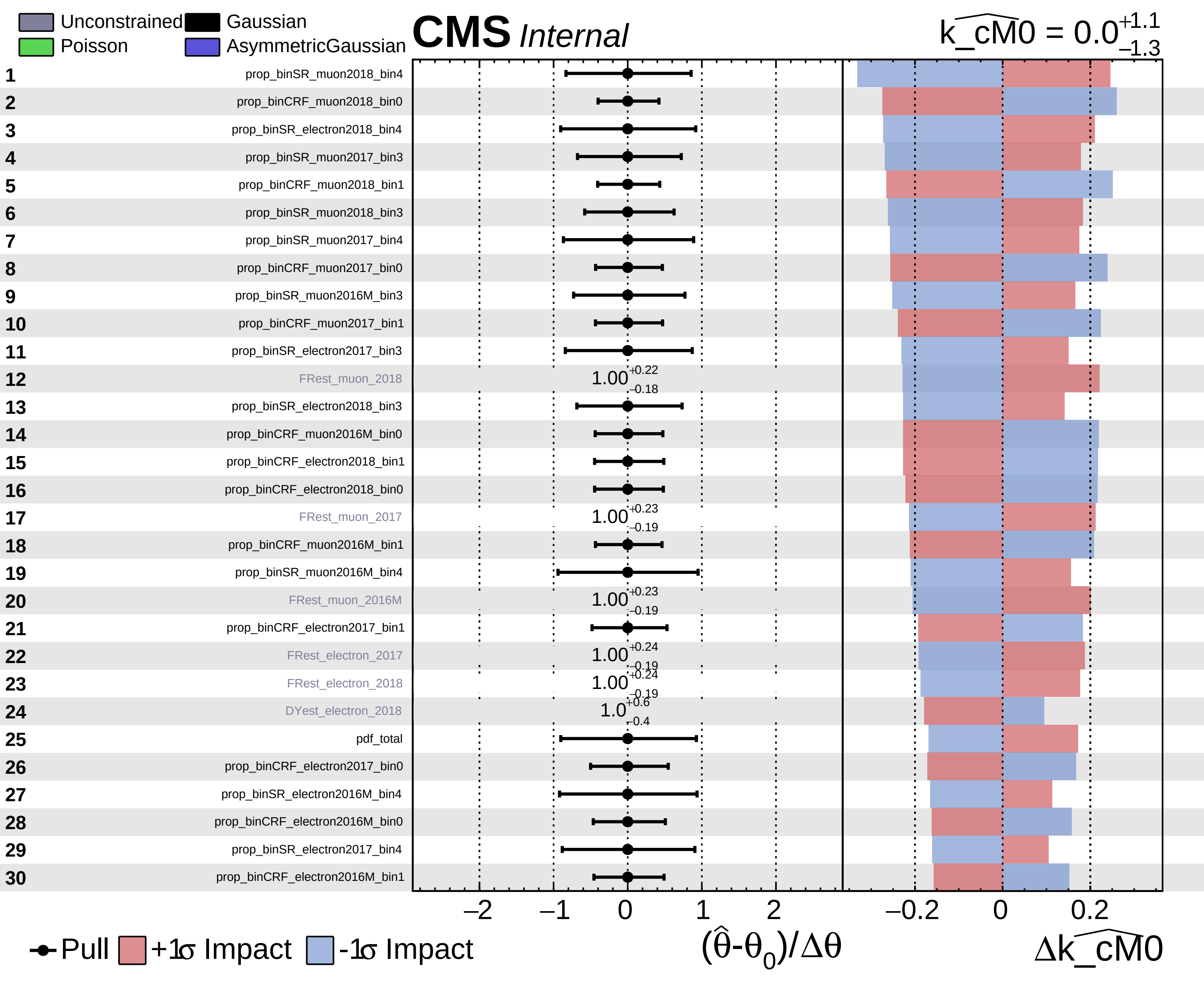
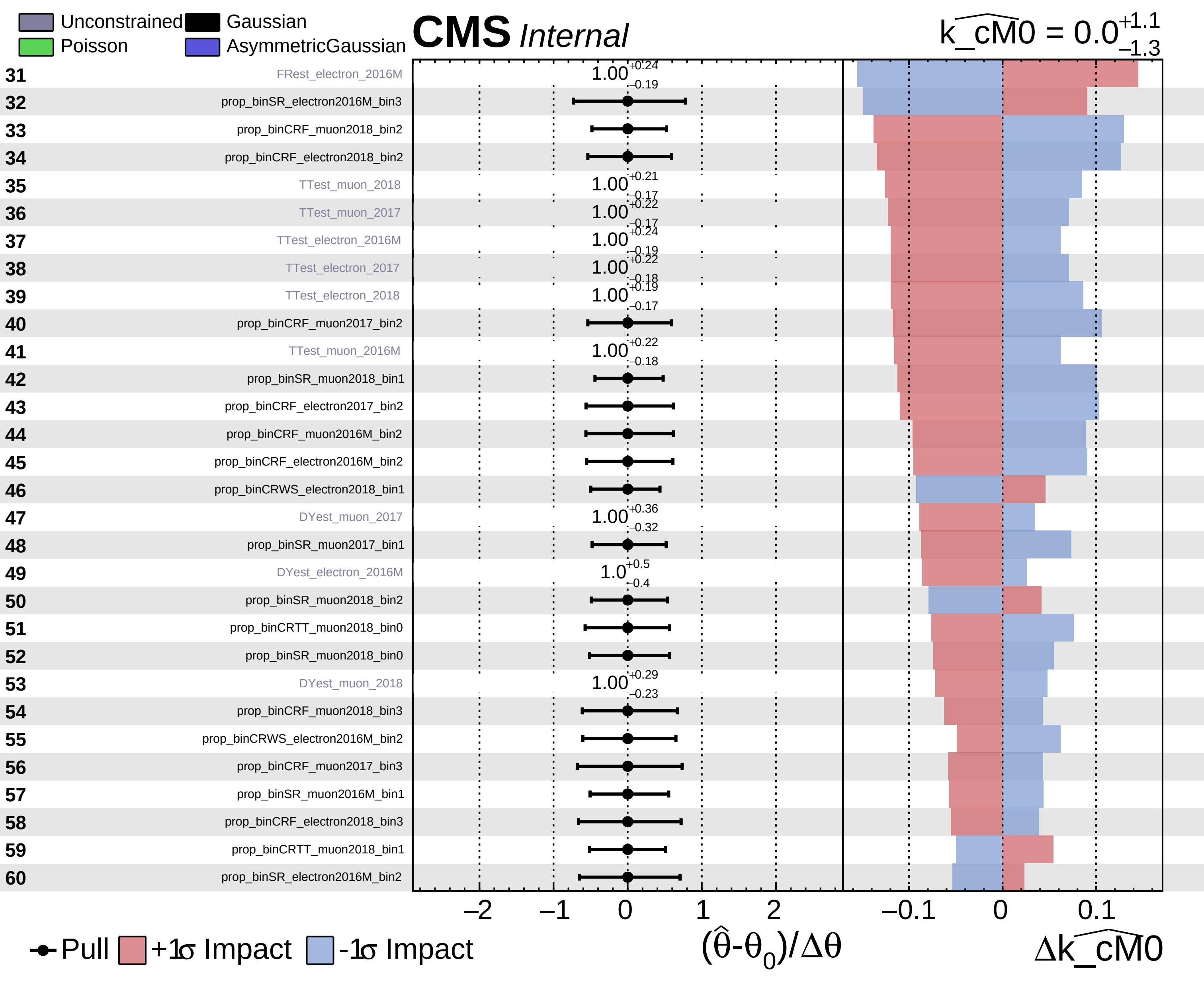
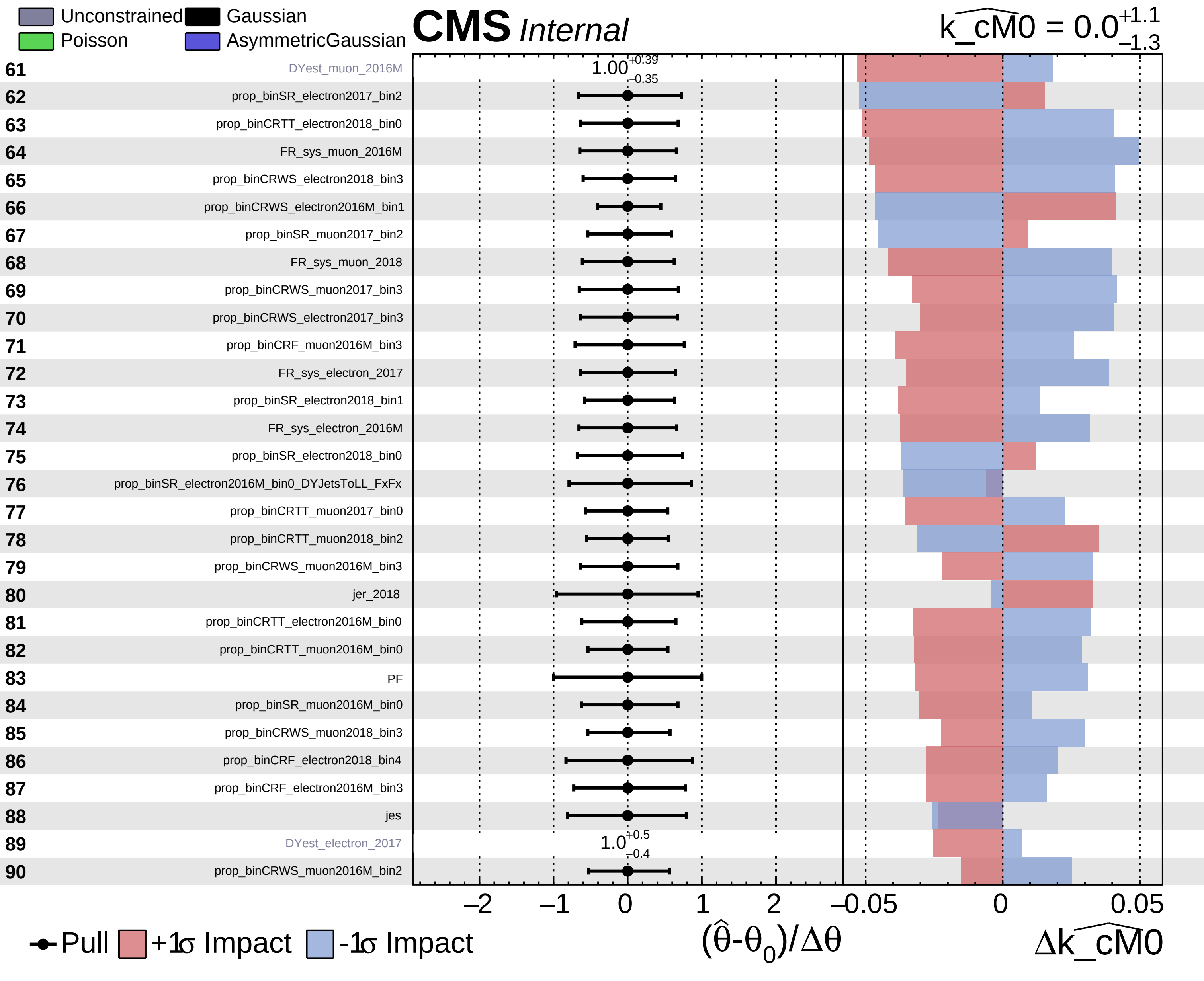
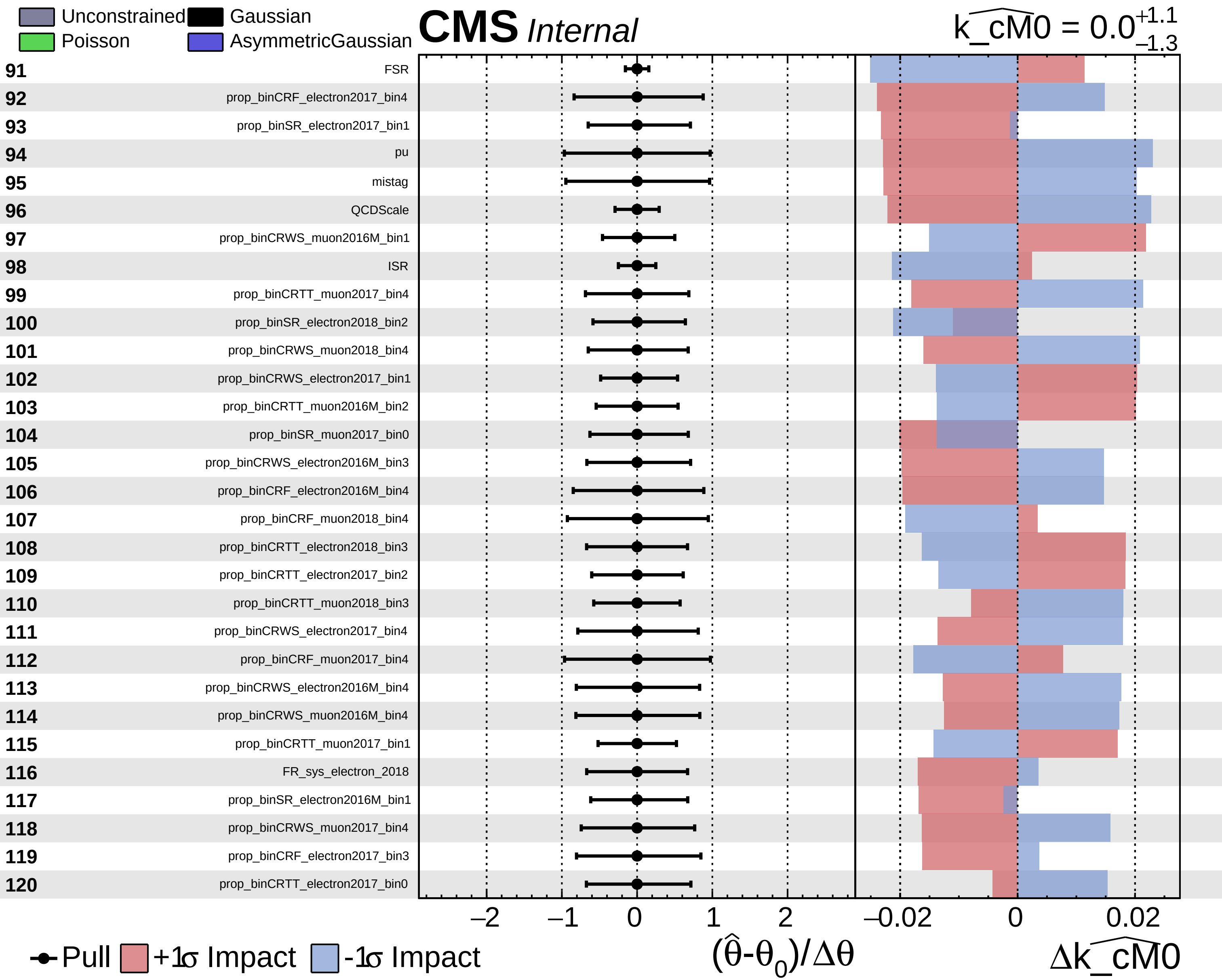

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+1.1}_{-1.3}$
1 prop_binSR_muon2018_bin4
2 prop_binCRF_muon2018_bin0
3 prop_binSR_electron2018_bin4
4 prop_binSR_muon2017_bin3
5 prop_binCRF_muon2018_bin1
6 prop_binSR_muon2018_bin3
7 prop_binSR_muon2017_bin4
8 prop_binCRF_muon2017_bin0
9 prop_binSR_muon2016M_bin3
10 prop_binCRF_muon2017_bin1
11 prop_binSR_electron2017_bin3
12 FRest_muon_2018 $1.00^{+0.22}_{-0.18}$
13 prop_binSR_electron2018_bin3
14 prop_binCRF_muon2016M_bin0
15 prop_binCRF_electron2018_bin1
16 prop_binCRF_electron2018_bin0
17 FRest_muon_2017 $1.00^{+0.23}_{-0.19}$
18 prop_binCRF_muon2016M_bin1
19 prop_binSR_muon2016M_bin4
20 FRest_muon_2016M $1.00^{+0.23}_{-0.19}$
21 prop_binCRF_electron2017_bin1
22 FRest_electron_2017 $1.00^{+0.24}_{-0.19}$
23 FRest_electron_2018 $1.00^{+0.24}_{-0.19}$
24 DYest_electron_2018 $1.0^{+0.6}_{-0.4}$
25 pdf_total
26 prop_binCRF_electron2017_bin0
27 prop_binSR_electron2016M_bin4
28 prop_binCRF_electron2016M_bin0
29 prop_binSR_electron2017_bin4
30 prop_binCRF_electron2016M_bin1
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

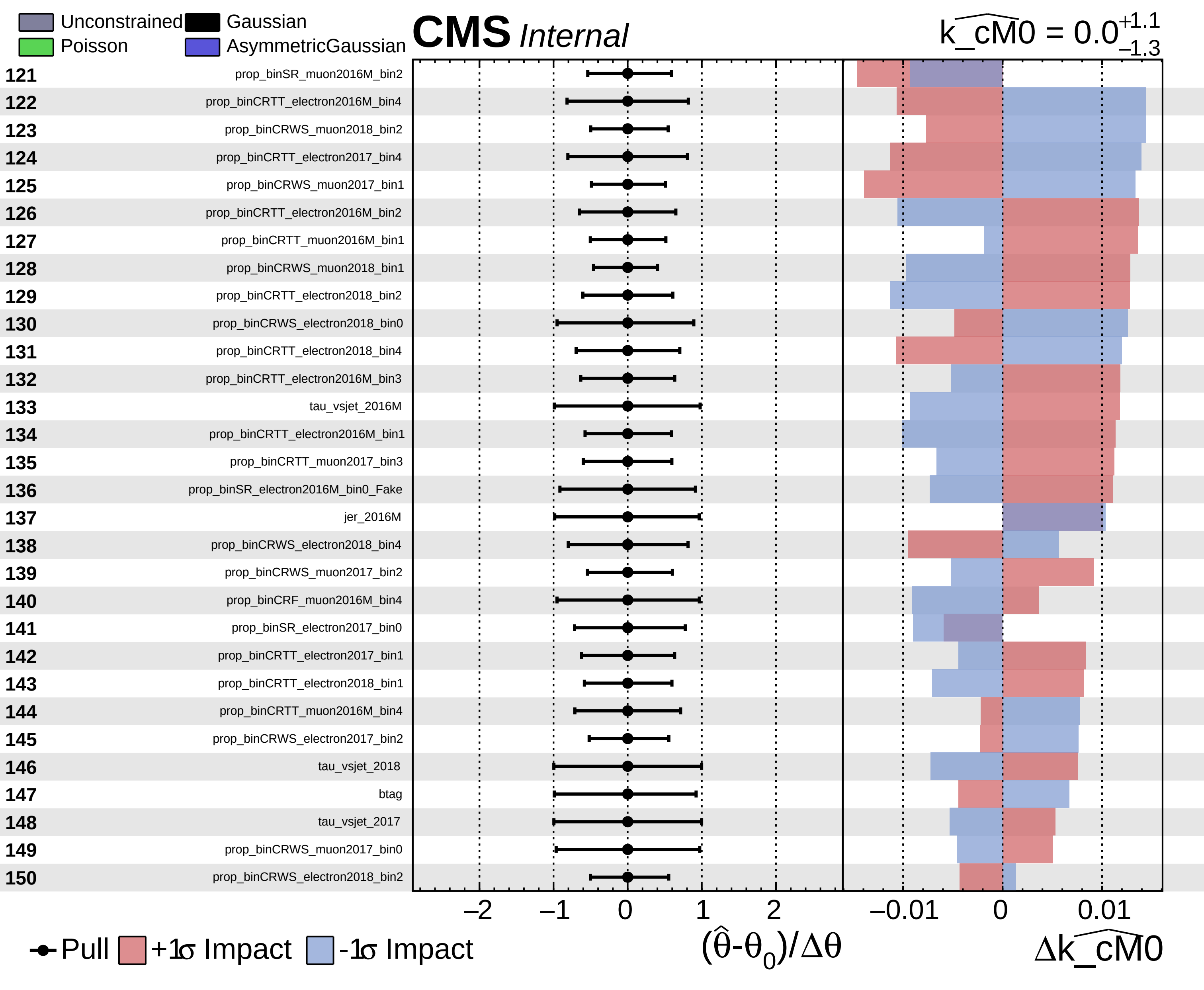


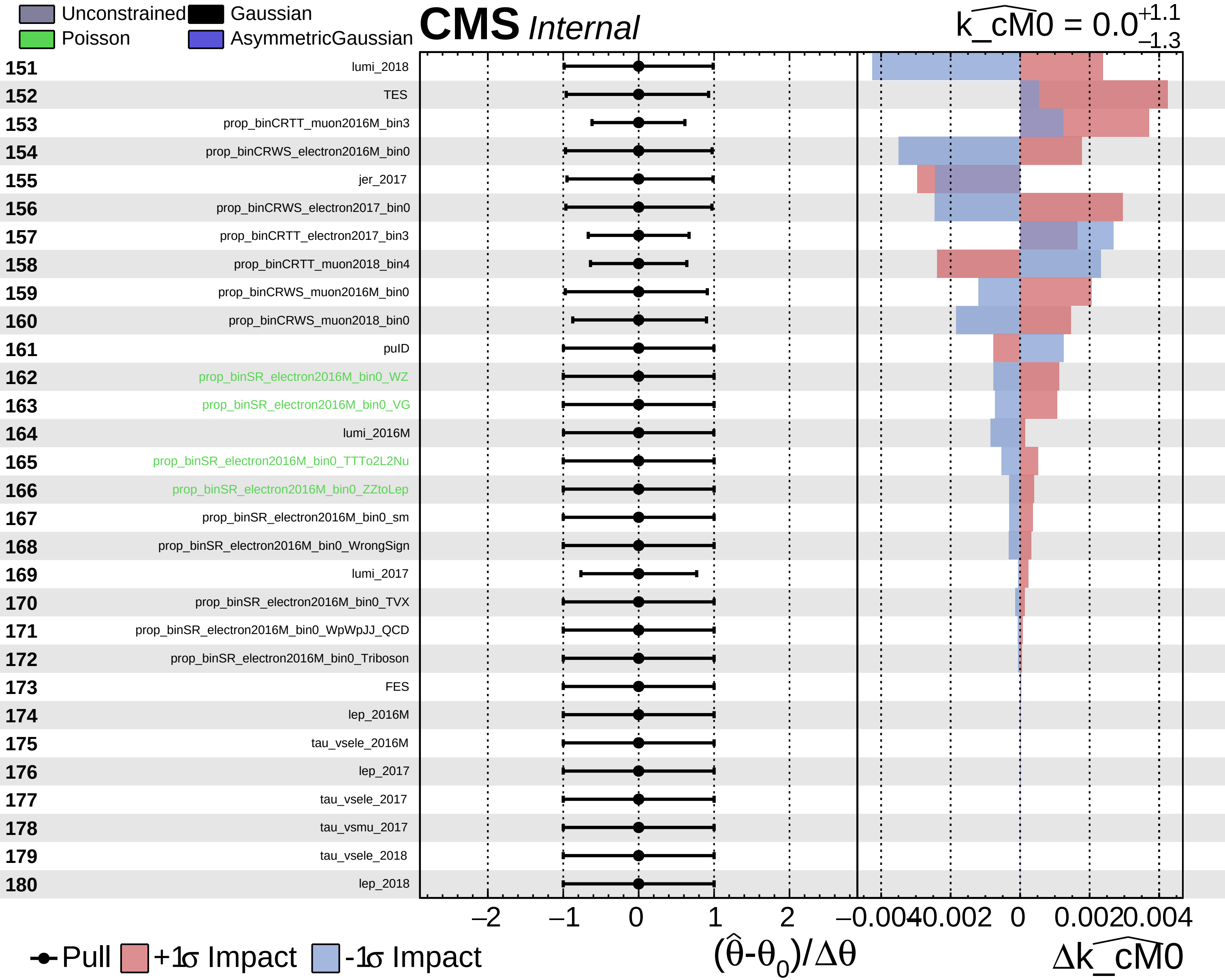
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+1.1}_{-1.3}$
31 FRest_electron_2016M
1.00 +0.24 −0.19
32 prop_binSR_electron2016M_bin3
33 prop_binCRF_muon2018_bin2
34 prop_binCRF_electron2018_bin2
35 TTest_muon_2018
1.00 +0.21 −0.17
36 TTest_muon_2017
1.00 +0.22 −0.17
37 TTest_electron_2016M
1.00 +0.24 −0.19
38 TTest_electron_2017
1.00 +0.22 −0.18
39 TTest_electron_2018
1.00 +0.19 −0.17
40 prop_binCRF_muon2017_bin2
41 TTest_muon_2016M
1.00 +0.22 −0.18
42 prop_binSR_muon2018_bin1
43 prop_binCRF_electron2017_bin2
44 prop_binCRF_muon2016M_bin2
45 prop_binCRF_electron2016M_bin2
46 prop_binCRWS_electron2018_bin1
47 DYest_muon_2017
1.00 +0.36 −0.32
48 prop_binSR_muon2017_bin1
49 DYest_electron_2016M
1.0 +0.5 −0.4
50 prop_binSR_muon2018_bin2
51 prop_binCRTT_muon2018_bin0
52 prop_binSR_muon2018_bin0
53 DYest_muon_2018
1.00 +0.29 −0.23
54 prop_binCRF_muon2018_bin3
55 prop_binCRWS_electron2016M_bin2
56 prop_binCRF_muon2017_bin3
57 prop_binSR_muon2016M_bin1
58 prop_binCRF_electron2018_bin3
59 prop_binCRTT_muon2018_bin1
60 prop_binSR_electron2016M_bin2
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



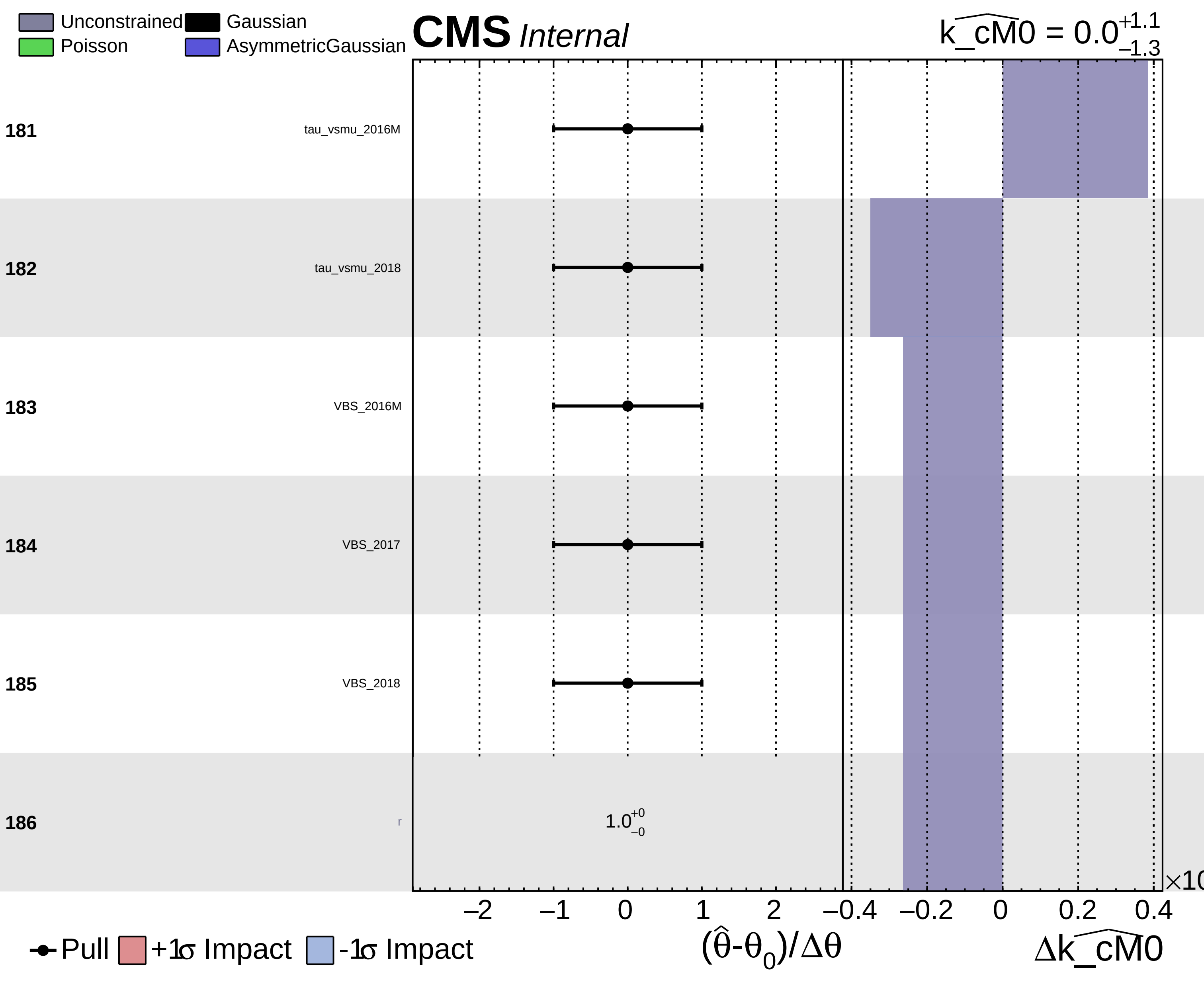
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +1.1 −1.3
61 DYest_muon_2016M 1.00 +0.39 −0.35
62 prop_binSR_electron2017_bin2
63 prop_binCRTT_electron2018_bin0
64 FR_sys_muon_2016M
65 prop_binCRWS_electron2018_bin3
66 prop_binCRWS_electron2016M_bin1
67 prop_binSR_muon2017_bin2
68 FR_sys_muon_2018
69 prop_binCRWS_muon2017_bin3
70 prop_binCRWS_electron2017_bin3
71 prop_binCRF_muon2016M_bin3
72 FR_sys_electron_2017
73 prop_binSR_electron2018_bin1
74 FR_sys_electron_2016M
75 prop_binSR_electron2018_bin0
76 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
77 prop_binCRTT_muon2017_bin0
78 prop_binCRTT_muon2018_bin2
79 prop_binCRWS_muon2016M_bin3
80 jer_2018
81 prop_binCRTT_electron2016M_bin0
82 prop_binCRTT_muon2016M_bin0
83 PF
84 prop_binSR_muon2016M_bin0
85 prop_binCRWS_muon2018_bin3
86 prop_binCRF_electron2018_bin4
87 prop_binCRF_electron2016M_bin3
88 jes
89 DYest_electron_2017 1.0 +0.5 −0.4
90 prop_binCRWS_muon2016M_bin2
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+1.1}_{-1.3}$
91 FSR
92 prop_binCRF_electron2017_bin4
93 prop_binSR_electron2017_bin1
94 pu
95 mistag
96 QCDScale
97 prop_binCRWS_muon2016M_bin1
98 ISR
99 prop_binCRTT_muon2017_bin4
100 prop_binSR_electron2018_bin2
101 prop_binCRWS_muon2018_bin4
102 prop_binCRWS_electron2017_bin1
103 prop_binCRTT_muon2016M_bin2
104 prop_binSR_muon2017_bin0
105 prop_binCRWS_electron2016M_bin3
106 prop_binCRF_electron2016M_bin4
107 prop_binCRF_muon2018_bin4
108 prop_binCRTT_electron2018_bin3
109 prop_binCRTT_electron2017_bin2
110 prop_binCRTT_muon2018_bin3
111 prop_binCRWS_electron2017_bin4
112 prop_binCRF_muon2017_bin4
113 prop_binCRWS_electron2016M_bin4
114 prop_binCRWS_muon2016M_bin4
115 prop_binCRTT_muon2017_bin1
116 FR_sys_electron_2018
117 prop_binSR_electron2016M_bin1
118 prop_binCRWS_muon2017_bin4
119 prop_binCRF_electron2017_bin3
120 prop_binCRTT_electron2017_bin0
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +1.1 −1.3
121 prop_binSR_muon2016M_bin2
122 prop_binCRTT_electron2016M_bin4
123 prop_binCRWS_muon2018_bin2
124 prop_binCRTT_electron2017_bin4
125 prop_binCRWS_muon2017_bin1
126 prop_binCRTT_electron2016M_bin2
127 prop_binCRTT_muon2016M_bin1
128 prop_binCRWS_muon2018_bin1
129 prop_binCRTT_electron2018_bin2
130 prop_binCRWS_electron2018_bin0
131 prop_binCRTT_electron2018_bin4
132 prop_binCRTT_electron2016M_bin3
133 tau_vsjet_2016M
134 prop_binCRTT_electron2016M_bin1
135 prop_binCRTT_muon2017_bin3
136 prop_binSR_electron2016M_bin0_Fake
137 jer_2016M
138 prop_binCRWS_electron2018_bin4
139 prop_binCRWS_muon2017_bin2
140 prop_binCRF_muon2016M_bin4
141 prop_binSR_electron2017_bin0
142 prop_binCRTT_electron2017_bin1
143 prop_binCRTT_electron2018_bin1
144 prop_binCRTT_muon2016M_bin4
145 prop_binCRWS_electron2017_bin2
146 tau_vsjet_2018
147 btag
148 tau_vsjet_2017
149 prop_binCRWS_muon2017_bin0
150 prop_binCRWS_electron2018_bin2
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+1.1}_{-1.3}$
151 lumi_2018
152 TES
153 prop_binCRTT_muon2016M_bin3
154 prop_binCRWS_electron2016M_bin0
155 jer_2017
156 prop_binCRWS_electron2017_bin0
157 prop_binCRTT_electron2017_bin3
158 prop_binCRTT_muon2018_bin4
159 prop_binCRWS_muon2016M_bin0
160 prop_binCRWS_muon2018_bin0
161 puID
162 prop_binSR_electron2016M_bin0_WZ
163 prop_binSR_electron2016M_bin0_VG
164 lumi_2016M
165 prop_binSR_electron2016M_bin0_TTTo2L2Nu
166 prop_binSR_electron2016M_bin0_ZZtoLep
167 prop_binSR_electron2016M_bin0_sm
168 prop_binSR_electron2016M_bin0_WrongSign
169 lumi_2017
170 prop_binSR_electron2016M_bin0_TVX
171 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
172 prop_binSR_electron2016M_bin0_Triboson
173 FES
174 lep_2016M
175 tau_vsele_2016M
176 lep_2017
177 tau_vsele_2017
178 tau_vsmu_2017
179 tau_vsele_2018
180 lep_2018
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +1.1 −1.3
181
tau_vsmu_2016M
182
tau_vsmu_2018
183
VBS_2016M
184
VBS_2017
185
VBS_2018
186
r
1.0 +0 −0
−2
−1
0
1
2
−0.4
−0.2
0
0.2
0.4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0